

From Transparency to Trust: A Literature Review on Explainable AI in Educational Systems

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Abstract: As artificial intelligence systems increasingly shape educational experiences, the demand for transparency and trust in algorithmic decision-making has led to a growing interest in the concept of Explainable AI (XAI). This literature review examines how XAI is being implemented in instructional contexts across different countries, highlighting both its pedagogical potentials and ethical tensions. Drawing from real-world applications and student reflections, this paper explores how explainability enhances feedback systems, supports learner autonomy, and fosters instructional alignment. However, paper also identifies persistent challenges, including cognitive over-reliance, educator deskilling, and the illusion of fairness. Through a series of international case studies, this review offers a comprehensive analysis of design considerations, policy frameworks, and user-centered practices for XAI in education. The paper concludes with a detailed checklist for ethical deployment and a set of forward-looking recommendations for ensuring that Explainable AI contributes meaningfully to equitable and human-centered learning environments.

Keywords: Explainable artificial intelligence, XAI, ethical AI, education, student-centered design.

Şeffaflıktan Güvene: Eğitim Sistemlerinde Açıklanabilir Yapay Zeka Üzerine Bir Literatür İncelemesi

Öz: Yapay zekâ (YZ) sistemleri eğitim deneyimlerini giderek daha fazla şekillendirdikçe, algoritmik karar alma süreçlerinde şeffaflık ve güvene olan talep, Açıklanabilir Yapay Zeka'ya (YZ) olan ilginin artmasına yol açmıştır. Bu literatür incelemesi, YZ'nin farklı ülkelerdeki eğitim ortamlarında nasıl uygulandığını inceleyerek hem pedagojik vaatlerini hem de etik gerilimlerini vurgulamaktadır. Gerçek dünya uygulamalarından ve öğrenci görüşlerinden yola çıkan çalışma, açıklanabilirliğin geri bildirim sistemlerini nasıl geliştirdiğini, öğrenci özerkliğini nasıl desteklediğini ve öğretimsel uyumu nasıl teşvik ettiğini araştırmaktadır. Bununla birlikte, bilişsel aşırı bağımlılık, eğitimci yetersizliği ve adalet yanılsaması gibi kalıcı zorlukları da belirlemektedir. Küresel vaka çalışmalarının bir sentezi aracılığıyla bu inceleme, eğitimde YZ için tasarım hususları, politika çerçeveleri ve kullanıcı merkezli uygulamalar hakkında kapsamlı bir analiz sunmaktadır. Çalışma, etik dağıtım için ayrıntılı bir kontrol listesi ve Açıklanabilir Yapay Zeka'nın eşitlikçi, insan merkezli öğrenme ortamlarına anlamlı bir şekilde katkıda bulunmasını sağlamayı amaçlayan bir dizi ileriye dönük öneriyle sonuçlanmaktadır.

Anahtar Kelimeler: Açıklanabilir yapay zekâ, XAI, etik yapay zekâ, eğitim, öğrenci merkezli tasarım.

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Introduction

Artificial Intelligence has become a transformative force in education, influencing all from automated grading and adaptive learning platforms to career guidance and student performance prediction (Islam et al., 2025). As artificial intelligence (AI) systems accept increasingly delicate roles in educational settings, concerns surrounding their opacity, bias, and lack of accountability have intensified. These concerns are particularly pressing in high-stakes contexts such as university admissions, essay scoring, and personalized learning recommendations where algorithmic decisions can significantly shape learners' futures.

To address these concerns, the concept of Explainable Artificial Intelligence (XAI) has emerged as a critical area of research and development. XAI refers to a set of methods and principles designed to make AI systems more transparent, interpretable, and accountable to human users (IBM, 2023). In education, the adoption of XAI is not simply a technical enhancement but a pedagogical, ethical, and policy imperative. Educational stakeholders (learners, educators, parents, and managers) increasingly demand to comprehend how and why AI systems arrive at specific outputs whether that could be a grade, a course recommendation, or a predicted dropout risk.

Recent literature suggests that explainability in educational AI is closely tied to broader values such as fairness (Deck et al., 2024), trust (Khosravi et al., 2022), and learner autonomy and inclusivity (Shafik, 2024). While the potential benefits of AI in education (ranging from real-time feedback to personalized pathways) are well recognized and become more visible to educational stakeholders, the risks of (un)intended discrimination, data misuse, and over-reliance on opaque systems have provoked scholars and practitioners to seek more interpretable alternatives.

Zhu and Wan Hussain (2025) examine AI awareness studies from 2019 to 2025 and presented that ethical vulnerabilities and job security concerns are dominant themes in the literature; these findings support the importance of transparent disclosure designs that take stakeholder concerns seriously in educational XAI systems.

This paper provides a literature review on the current applications, challenges, and global cases of XAI in education. Drawing from empirical examples in diverse educational systems - including those in the United States, China, India, South Korea, Brazil, and Europe - it synthesizes how XAI is being implemented in real-world educational contexts and what implications arise for policy, practice, and further research.

The main aim of this paper is to plan the educational significance of explainability by exploring both technical solutions and human-centered strategies. In doing so, this review contributes to ongoing conversations about how to balance innovation with transparency, equity, and educational integrity.

Theoretical Background

Explainable Artificial Intelligence (XAI) refers to a set of methods and design principles that make the outputs and inner workings of AI systems understandable to human users (Doshi-Velez & Kim, 2017). In contrast to traditional 'black-box' models where internal mechanisms are hidden to the users - such as in the case of deep learning's neural networks - XAI techniques aim to provide transparency by revealing how an algorithm arrived at a specific decision, what features or variables influenced it, and how confident the system is in its output (Samek et al., 2017; Patil & Hurix Digital, 2024). The goal is not only to foster technical auditability but to enable meaningful human interpretation and oversight, particularly in high-stakes settings such as education, healthcare, financial services and criminal justice (IBM, 2023).

In educational contexts, explainability takes on unique significance. Learning is a socially and cognitively mediated process, where feedback, reflection, and justification play a critical role in knowledge acquisition and learner development (Butler & Winne, 1995; Vygotsky, 1978). From the perspective of cognitive learning theory, for example, students internalize new information more effectively when they can trace the reasoning behind feedback and link it to prior knowledge (Bransford et al., 2000). Similarly, in constructivist pedagogy, learners actively construct meaning through engagement with explanations, reasoning, and justifications - highlighting the importance of transparent instructional interventions (Vygotsky, 1978). XAI, when well implemented, aligns with these theories by allowing AI systems to serve not just as automated tools but as partners in the learning process that metacognition (Winne & Hadwin, 1998), self-regulation (Zimmerman, 2002), and critical thinking (Facione, 1990).

The educational relevance of XAI is further grounded in AI ethics and digital trust frameworks, which emphasize the necessity of interpretability, fairness, and accountability. According to the European Commission's High-Level Expert Group on AI (2019), explainability is one of the key pillars of trustworthy AI. Similarly, UNESCO's (2021) Recommendation on the Ethics of Artificial Intelligence highlights the need for transparency and intelligibility in AI applications, especially those affecting children and youth. In the context of schooling, where students may not have the capacity or authority to challenge AI decisions, explainability serves as a defense against harm, ensuring that AI enhanced tools remain subject to study and aligned with human values.

In this regard, Kong and Zhu (2025) developed and validated a multidimensional AI Ethical Awareness Scale (AIEAS) within a problem-solving-based AI literacy course, demonstrating that students' ethical sensitivity can be systematically measured and used to inform the design of AI-supported learning environments. The results confirmed that the three-factor structure of the AIEAS (human autonomy, beneficence, and justice) have strong validity and reliability. Learning activities grounded in the scale, together with reflective writing tasks, significantly enhanced students' ethical awareness. Overall, the study shows that cultivating AI ethical awareness through problem-solving with AI tools is both feasible and pedagogically effective, offering a useful reference point for XAI designs that explicitly aim to foreground ethical dimensions in educational contexts.

Technically, explainability can be achieved through a variety of approaches. Post-hoc techniques such as SHAP (Shapley Additive Explanations) (Lundberg, 2018), LIME (Local Interpretable Model-agnostic Explanations) (Papers with Code, n.d.), and saliency maps (Subhash, 2022) provide insight into which features influenced a given output, often visualized as heatmaps, weights, or textual rationales (Lundberg & Lee, 2017; Ribeiro, Singh & Guestrin, 2016). For example, figure 1 presents a saliency map that highlights the pixels most relevant for correctly classifying the images on the left (GeeksforGeeks, 2021).

Figure 1.

The Image on the Right Represents the Saliency Map Corresponding to the One on the Left



In contrast, inherently interpretable models (such as decision trees, rule-based classifiers, or sparse linear models) embed transparency in their structure, though often at the cost of predictive performance. According to Holstein et al. (2019), in education, a hybrid approach is often necessary by combining high-performing AI models with user-facing interfaces that translate model behavior into educationally meaningful language for students, teachers, and administrators.

The current literature also distinguishes between global and local explainability. Global explanations provide an overview of how a model works overall - useful for developers and policy stakeholders - while local explanations describe why a specific output was given for a particular input, which is especially relevant for end-users like students and

instructors. For example, an adaptive learning system may use global explainability to justify its content sequencing rules, and local explainability to inform a student why they received a specific recommendation or score. In both cases, the goal is to support human-centered AI systems that communicate their logic clearly and foster trust and agency among users (Zhou, Bischl & Molnar, 2021).

Taken together, these theoretical perspectives drawn from education, ethics, and computer science, define XAI more than a technical challenge. It is a multi-dimensional approach that bridges pedagogy, policy, and human-computer interaction. In the following sections, the author applies this lens to real-world educational systems and tools, examining how XAI principles are implemented in practice across global contexts and what insights can be drawn from their successes and limitations.

Applications of Explainable AI in Education

Explainable Artificial Intelligence has rapidly moved from theory to practice in educational technology, transforming how learning is delivered, assessed, and personalized (Chen et al., 2021; Holstein et al., 2019). As artificial intelligence becomes more embedded in classroom platforms and administrative systems, explainability is increasingly recognized as essential to ensure that both students and educators understand, trust, and effectively interact with AI-driven outputs (Khosravi et al., 2022). Without transparency, users may misinterpret or over-rely on algorithmic feedback, potentially undermining pedagogical goals and learner autonomy (Chamola et al., 2023).

This section explores the key areas where XAI is being applied in education, with a focus on automated assessment, adaptive learning, real-time feedback, and educational decision-making systems. Across each domain, XAI facilitates transparency, enhances learner engagement, and supports fairness and pedagogical alignment.

Automated Essay Scoring

One of the earliest and most widely deployed AI applications in education is Automated Essay Scoring (AES). These systems evaluate written responses using machine learning models trained on large amounts of human-scored essays. However, the opacity of such systems has often been a source of controversy. XAI enhances AES systems by providing rationale-based feedback, allowing learners and educators to understand which features, such as grammar, coherence, structural organization, vocabulary richness, and clarity of argumentation, contributed to a given score. This transparency not only increases trust in the scoring process but also supports formative learning by helping students reflect on specific aspects of their writing that influenced the outcome (Dadi & Sanampudi, 2021; Ke & Ng, 2021).

For instance, XAI-enabled AES could visually highlight sentence structures or word choices that positively or negatively affected the final grade, thereby promoting more effective revision strategies and metacognitive awareness. Moreover, by revealing the decision pathways of the underlying models, XAI helps reduce bias and ensures the fairness of assessment, particularly for linguistically diverse learners (Khosravi et al., 2022). Models like e-rater and Project Essay Grade (PEG) have been adapted with explanation layers that display annotated feedback or scoring rubrics linked to essay segments (Shermis & Burstein, 2013).

More recent models incorporate SHAP (Shapley Additive Explanations) and attention-based visualizations to allow students and teachers to see *why* certain aspects of an essay were flagged for improvement. Students report feeling more motivated and confident when they can see a breakdown of their score and understand how to revise effectively which is an observation echoed in multiple student projects that emphasized the value of actionable AI-generated writing feedback. Teachers also benefit from explanation dashboards that help validate the fairness of scores or identify problematic patterns, such as penalizing essays with non-standard English (Alkhatlan & Kalita, 2019).

Adaptive Learning Systems

Artificial intelligence influences adaptive learning systems to personalize instructional content in real time, dynamically adjusting the sequence, pace, and difficulty of materials based on each learner's performance and interaction patterns. XAI plays a crucial role in these systems by revealing the rationale behind such adaptations, making the learning process more transparent and trustworthy (Holstein et al., 2019; Khosravi et al., 2022). Islam et al. (2025) discuss the integration of XAI into educational data mining, demonstrating how support systems can be made more transparent and actionable along with student achievement prediction.

Without explainability, learners may feel confused, disoriented, or even disempowered when re-directed to remedial exercises or presented with unexpectedly challenging material, which can adversely impact engagement and motivation. XAI helps mitigate these issues by providing interpretable feedback that supports metacognitive development and learner autonomy.

Platforms such as Knewton, Smart Sparrow, and ALEKS (Table 1) have integrated explainable components, such as skill maps, progress justifications, and 'why this recommendation' buttons to communicate the basis of adaptive

decisions (Holstein et al., 2020). By exposing the system's reasoning (e.g., "You missed two conceptually linked questions, so we recommend reviewing Topic X"), XAI aligns with educational theories that prioritize learner autonomy and reflection (Winne & Hadwin, 1998).

Table 1.

Comparison of Adaptive Learning Platforms: Knewton, Smart Sparrow, and ALEKS

Feature / Platform	Knewton	Smart Sparrow	ALEKS
Developer / Owner	Pearson (formerly independent)	Pearson (acquired 2018)	McGraw Hill Education
Primary Subject Focus	Math, science, business	STEM, medical education, custom courses	Mathematics, chemistry, accounting
Target Learners	Higher education, K–12	Higher education, vocational training	K–12, college, adult learners
Adaptive Learning Method	Real-time personalization using learning analytics	Learner-driven pathways with branching logic	AI-based knowledge space theory with dynamic assessment
Use of AI	Yes – predicts performance, recommends content	Yes – rule-based AI in lesson design	Yes – determines mastery through ongoing assessment
Explainability (XAI)	Limited transparency to students and instructors	Some transparency via feedback and branching	Moderate – progress tracking and topic mastery shown
Customization	Low for instructors; prebuilt by Pearson	High – instructors build and control content	Low – system-generated pathways
Strengths	Scalable, aligned with Pearson content	Instructor control, effective for simulations	Strong diagnostics, mastery-based progression
Limitations	Limited adaptation transparency	Complex setup for educators	Limited feedback and customization for learners

Note. Content is drawn from Holstein et al. (2020), Khosravi et al. (2022) and Yudelson et al. (2013).

Real-Time Feedback and Language Learning

In language and communication education, XAI enhances AI-powered feedback tools by providing interpretable and pedagogically aligned corrections. For example, pronunciation tools such as ELSA Speak, SpeakBuddy, or iFLYTEK TalkPal (Table 2) have moved beyond raw scoring to offer visualizations (e.g., sound wave comparisons, phoneme highlights) and textual explanations ("Your tongue placement was incorrect for the /θ/ sound"). These explainable elements not only support phonological awareness but also reduce learner anxiety by framing AI as a helpful guide rather than an opaque evaluator (Akhmad & Munawir, 2022; Kholis, 2021).

Table 2.

Comparison of Pronunciation Tools: ELSA Speak, SpeakBuddy, and iFLYTEK TalkPal

Feature / Tool	ELSA Speak	SpeakBuddy	iFLYTEK TalkPal
Developer / Owner	Elsa Corp	Xinhui Technology Co., Ltd.	iFLYTEK (China-based AI company)
Core Functionality	AI-powered pronunciation feedback with phonetic scoring and instant corrections	AI-driven interactive oral practice across life scenarios	GPT-based tutor covering speaking, listening, writing
Feedback Method	Immediate visual & auditory comparison to target pronunciation; phonetic guidance	Intelligent error correction within scenario dialogues	Real-time pronunciation feedback embedded in conversational practice
Supported Languages	English (US accent focus)	English (scenario-based only)	50+ languages, including English, Chinese, etc.
Pedagogical Features	Structured phonetic curriculum; gamified lessons; lesson themes	Multiple scenarios: daily life, business; freeform speak practice	Role-play, debates, call mode, grammar integrated practice

Feature / Tool	ELSA Speak	SpeakBuddy	iFLYTEK TalkPal
Target Learners	EFL learners, formal/informal educational settings	Casual to advanced learners seeking conversation fluency	Broad language learners across proficiency levels
Strengths	Robust instant corrective feedback; well-researched success in pronunciation gains	Immersive scenario-based dialogue; accessible anytime	Multimodal skills support; AI-driven customization
Limitations	Free version limited; premium needed for full content; less focus on supra-segments	Limited to English; less research-backed efficacy data	Fewer peer-reviewed studies; depends on internet connection

Note. Table synthesizes information from app records and research (e.g. Anggraini, 2022; Sholekhah & Fakhurriana, 2023; Xinhui Technology, 2025)

Students report greater satisfaction and learning gains when language feedback includes rule-based or grammar-enhanced explanations (e.g. “Passive voice was used incorrectly due to verb agreement”). This aligns with cognitive load theory, which emphasizes the importance of structured, meaningful feedback in second language acquisition (Sweller, Ayres & Kalyuga, 2011). Furthermore, multilingual explainability can improve inclusivity and access, particularly for learners in linguistically diverse settings.

Student Performance Prediction and At-Risk Alerts

AI-driven learning analytics are increasingly employed in higher education to identify students at risk of academic failure, withdrawal, or disengagement. These systems use a variety of indicators, such as declining grades, reduced platform activity, and missed deadlines to flag students who may benefit from early intervention. However, without transparency in how these determinations are made, such predictive tools risk being perceived as intrusive, opaque, or even unfair.

This is where XAI becomes essential. By ensuring that predictions are not only data-driven but also interpretable, XAI enhances the credibility and educational value of early-warning systems. For example, tools that incorporate decision trees or SHAP visualizations allow instructors to understand the specific features or behaviors such as low quiz scores, inconsistent attendance, or a sudden drop in online engagement that triggered an at-risk classification. This level of explainability supports both accountability and instructional action, enabling educators to make informed decisions rather than blindly following algorithmic prompts.

Empirical research supports the importance of such transparency. Studies in higher education contexts have demonstrated that teacher trust in predictive alerts increases when explanation mechanisms are embedded. Feldman-Maggor et al. (2025), for instance, found that chemistry teachers interacting with AI-based recommendation tools expressed significantly greater acceptance when domain-specific explanations were included alongside general analytics. This combination not only built trust but also encouraged teachers to engage more critically and constructively with the AI outputs.

Explainable prediction systems are most effective when integrated into a responsive human–AI feedback loop. Rather than operating as standalone decision-making tools, XAI-enhanced early warning systems can prompt timely and targeted human interventions, such as counselor outreach, personalized instructor feedback, or the recommendation of supplemental learning resources. By making the rationale behind predictions visible and actionable, these systems foster alignment between algorithmic insights and pedagogical support. For instance, Embarak (2025) proposes an explainable personalization framework integrated with Internet of Behavior data, offering a behavior-based intelligent training approach that makes visible not only what is recommended but also why.

For instance, SHAP-based analytics can identify the most significant learning gaps and suggest personalized content to address them, empowering students to take informed steps toward improvement (Hur et al., 2022). Likewise, interpretable dashboards visualizing the behavioral indicators contributing to dropout risk enable instructors to offer evidence-based interventions, reinforcing trust in the system and enhancing its practical value in educational settings (Nagy & Molontay, 2024). This synergy between human judgment and algorithmic guidance exemplifies a responsible and effective model for using AI in student success initiatives. Finally, Johora et al. (2025) developed a model that predicts academic achievement of undergraduate students with SHAP and LIME-based explanations, demonstrating that XAI can be a tool to enhance both performance and educator confidence.

AI in Educational Counseling and Career Guidance

Emerging AI applications in educational counseling such as recommending career tracks, university programs, or elective courses highlight the critical role of value-sensitive explainability. Students are more likely to trust and act on guidance when they understand its rationale, for instance, ‘You were matched with environmental science due to your interests in biology, climate change, and strong math performance.’

Platforms like Naviance and India’s CareerNeeti are beginning to integrate natural language explanations and interactive dashboards that align student profiles with recommended pathways, enhancing transparency and perceived relevance (PowerSchool, 2025; Tiwari, 2025).

In contrast, black-box recommendation systems can erode trust and amplify inequities, especially when students feel pigeonholed or misclassified without insight into the decision process (Zhang et al., 2021). Ethical concerns around such opaque systems have led student-led design projects to emphasize the need for interfaces supporting questioning, personalization, and informed consent ensuring learners can understand, challenge, and opt into AI-driven suggestions rather than being passively directed.

Table 3.

Comparison of AI-Powered Career Guidance Platforms: Naviance vs. CareerNeeti

Feature	Naviance	CareerNeeti
Developer / Ownership	PowerSchool (USA-based)	Clastify EdTech Pvt. Ltd. (India-based)
Target Users	K–12 students, counselors, parents	Higher secondary and undergraduate students
Primary Purpose	College, career, and life readiness planning	Personalized career discovery and academic guidance
Personalization	Uses GPA, test scores, interests, and career inventories to recommend college and career matches	AI-driven recommendations based on academic strengths, interests, aptitude, and career goals
Explainability Features	Scattergrams, SuperMatch, and visual metrics; limited natural language justification	Natural language feedback, dashboard-based rationales, and transparent rule-based logic
Counselor Tools	Student tracking, messaging, outcome analytics, and bulk processing features	Counselor-assisted interface with student profiles and skill alignment maps
Transparency	Moderate; visual tools explain outcomes, but internal AI logic remains mostly opaque	High; system explains why each suggestion was made using data and rule-based decision paths
Research-Backed Evidence	Widely used in U.S. districts; counselor satisfaction and usability studies exist	Early field trials in India; positive feedback on trust and decision-making from pilot studies
Potential Concerns	May feel impersonal or rigid; lacks full AI transparency	Emerging platform; needs more peer-reviewed validation for large-scale adoption

Note. Comparison is based on public product documentation, platform overviews, and academic reports available as of 2025.

Global Case Studies of XAI in Education

While the principles of XAI are widely acknowledged in academic and ethical discourse, their actual implementation in educational systems varies greatly across contexts. This section highlights international case studies that demonstrate how different countries and institutions have approached the design and deployment of XAI in education. These cases illuminate both the pedagogical benefits and the socio-technical challenges of making AI systems transparent and trustworthy in diverse educational settings.

The United States: Enhancing Transparency in Automated Grading and Feedback

In the United States, educational institutions have led the adoption of AI-powered grading systems and personalized feedback tools, especially in K–12 and higher education. Platforms such as Turnitin Draft Coach and Gradescope now include explainability features that help students better understand how their scores and feedback are generated. These features, such as inline comments, rubric-based breakdowns, and visual indicators, support formative assessment and foster academic integrity.

For example, Turnitin Draft Coach provides real-time, in-app feedback on grammar, citation, and overall structure, guiding students as they write and allowing for continuous revision. Its integration into platforms like Microsoft Word and Google Docs enables students to track and address writing issues before submission, which enhances their confidence and engagement with feedback (Turnitin Draft Coach, 2025). Similarly, Gradescope employs AI-assisted grading through a system of ‘answer groups,’ which clusters similar responses for instructor review. Educators can view and edit grouped feedback, allowing for consistency while also flagging uncertainty in borderline cases, thereby increasing grading transparency and instructional control (Gradescope Guides, 2024).

A particularly illustrative innovation is the pilot implementation of OpenAI’s GPT-4o in university writing centers to provide automated essay scoring and formative feedback. While GPT models can deliver quick and fluent responses, a common criticism is their ‘black-box’ nature, students often struggle to understand how scores or suggestions are derived. To mitigate this, some institutions have introduced confidence explanation panels, visual modules that highlight which parts of a student’s essay correspond to specific rubric criteria (e.g., coherence, grammar, argument strength). This strategy aligns with broader AI design principles that prioritize user trust and meaningful interaction (Karayev & Gutowski, 2020).

Preliminary student feedback suggests that these explanation-enhanced tools help demystify AI-generated assessments, enabling more targeted revisions and deeper reflection on writing quality. Collectively, these developments in explainable AI point toward a more transparent, user-centered approach to automated grading in U.S. educational settings.

The Netherlands: Explainability as a Driver of Trust

Dutch universities have actively piloted XAI features within formative assessment systems to foster transparency and trust. A notable study at the University of Twente explored how explainable visualizations in automated grading tools influence student trust and acceptance. Through interviews and mock-up testing, researchers evaluated multiple forms of explanations, including feature importance visuals, rubric-aligned feedback, and confidence indicators, and found that students were significantly more likely to trust and act on feedback when accompanied by clear, interpretable explanations (e.g., which errors impacted their score and why) (Stoian, 2023).

In one implementation, students received an AI-generated summary of their assignment, followed by ‘explanation cards’ that highlighted specific textual features, such as excessive passive voice or repetitive phrasing and showed how these factors affected their performance. These cards were co-designed with linguistics faculty to ensure pedagogical relevance. Students reported that this enhanced clarity improved engagement and reduced resistance to machine-generated feedback, while educators appreciated the ability to easily verify, override, or supplement AI decisions.

By grounding AI insights in human-understandable language and contextual criteria, this Dutch initiative exemplifies how XAI can strengthen formative assessment and it supports learner agency, promotes reflective revision, and ensures that algorithmic support complements, rather than replaces, teacher judgment.

The United Kingdom: Explainability and Student Autonomy in Writing Support

In the United Kingdom, universities have increasingly integrated AI-powered tools like Turnitin Draft Coach to support formative academic writing. Draft Coach offers real-time, in-document feedback on grammar, citations, and text similarity, paired with concise, explainable guidance. For example, “This sentence uses passive voice, which may reduce clarity,” or “This paragraph lacks citation despite referencing a known idea.” Such feedback is designed not to prescribe solutions but to guide student reflection and decision-making.

Case evidence from institutions such as Plymouth University indicates that students benefit from these explainable elements during the drafting process, allowing them to revise independently before submitting work for formal assessment. Educators report that Draft Coach reduces their workload during early feedback stages, enabling them to focus on higher-order concerns like argument quality and structure (Plymouth University, 2022; Turnitin, 2023). This aligns with findings from broader studies on AI-assisted writing tools: explainable, rule-based feedback systems help students better understand errors and apply metacognitive revision strategies, thus supporting autonomy rather than dependency (Nguyen et al., 2024).

Moreover, recent discourse in the UK higher education highlights concerns about opaque uses of AI in assessment. Surveys from the Higher Education Policy Institute (HEPI) suggest that students are more accepting of AI-powered feedback when it is transparent, skill-focused, and aligned with institutional writing values. Explainability, through interpretable comments, rubric alignment, and editable drafts, is increasingly seen not just as a technical feature, but as a pedagogical imperative (Neves, 2024).

Australia: “Mark My Words” and Personalized Feedback

In Australia, platforms like Mark My Words (developed with Monash University partners) exemplify how explainable AI can support formative writing feedback. The tool uses AI to detect structural, lexical, and argumentative issues in student essays, then offers feedback in clear, student-friendly language with confidence labels such as ‘likely off-topic,’ ‘moderately coherent,’ or ‘high clarity.’ Teachers and students value its transparency slider, which lets them control the depth of feedback displayed and understand the AI’s reasoning. Crucially, the tool includes mechanisms for student agency, most notably a ‘Disagree with AI’ button, empowering learners to challenge or reflect on flagged feedback. This feature reinforces critical engagement and trust, moving beyond passive acceptance of automated suggestions.

Supporting the pedagogical impact of such tools, Monash University researchers Jin et al. (2024) developed PolyFeed, a feedback analytics prototype aimed at scaffolding student feedback literacy. In their study, students co-designed features like annotation, to-do planning, and dashboard summaries, exactly the kind of interactive elements Mark My Words provides. They found that when learners actively interacted with feedback analytics, they displayed deeper reflection, improved satisfaction, and stronger engagement with their writing tasks, especially when they could track how comments mapped to learning actions.

China and South Korea: Large-Scale Adaptive Platforms

In East Asia, countries like China and South Korea have deployed large-scale AI-powered learning platforms, such as iFLYTEK Smart Learning, Squirrel AI, and Riiid’s Santa Tutor, to deliver personalized instruction tailored to learner performance, history, and behavior.

In China, iFLYTEK’s smart learning solutions are now operational in over 3,600 schools across 28 provinces, featuring AI-driven homework assignments, exam grading, and personalized learning maps based on student data (PR Newswire, 2022). While early versions provided little insight into recommendation logic, recent updates include explainability modules that justify content sequencing. For example, if a student struggles with quadratic transformations, the system explains why they are recommended remedial algebra exercises.

In South Korea, Riiid’s Santa Tutor, a widely recognized AI tutor for TOEIC and other standardized exams, has been upgraded to include XAI features. The platform now provides explanations for recommended learning paths, enabling teachers to understand and adjust AI-driven pacing to avoid holding back high-performing students (KoreaTechDesk, 2019). These explanatory additions typically include statements such as: “You struggled with quadratic transformations, which are a prerequisite for the next topic.” Teachers in both countries emphasize that when the system reveals its reasoning, alignment with classroom curricula improves and resistances from learners (especially high-achievers) are reduced (Riiid, 2024).

By embedding transparent rationale into AI recommendations, these platforms move toward a responsible educational model by reducing opacity in large-scale personalization and supporting co-learning between students, educators, and AI systems.

Brazil: Explainable Learning Analytics Dashboards

Brazilian universities are actively piloting learning analytics dashboards (LADs) equipped with XAI features to enhance educators’ capacity to monitor student engagement and performance in real time. These dashboards integrate predictive analytics, such as dropout risk scores with traceable feature breakdowns, highlighting factors like low quiz frequency or erratic login patterns that contribute to those risks.

Beyond simply displaying risk scores, these systems surface interpretable insights. For example, a teacher can see that a student’s low engagement derives from missing three quizzes and several irregular login events within the past week. These explainable indicators help teachers to contextualize data-driven alerts and craft interventions grounded in the students’ real behaviors. This contrasts with dashboards that merely present opaque risk flags, which can weaken trust and encourage overreliance on automated signals.

Empirical research supports this XAI-forward approach. A teacher-facing LAD developed at a Brazilian university, focused on online learning analytics, was co-designed with educators and iteratively tested (Dourado et al., 2021). The system enabled instructors to trace the process-oriented feedback loop, revealing patterns in submissions and login behavior, which educators used to identify disengagement and tailor support. Teachers reported greater confidence in the AI system and a stronger alignment between analytic signals and pedagogical intent, citing the transparent breakdown of features as critical to building trust.

Ethical Challenges and Limitations

Recent evidence shows that ethical gaps persist even in AI-supported educational tools beyond academic prediction or assessment. For example, Fütterer et al. (2025) reviewed 104 classroom management studies and found that AI solutions for participation tracking, attention monitoring, and behavior analysis are rapidly proliferating, yet ethical and

privacy implications remain largely unaddressed. Their findings indicate that explainability alone is insufficient; without robust ethical safeguards, XAI cannot fully mitigate risks related to surveillance, data misuse, and inequitable treatment of students.

Despite the growing implementation of XAI in education, significant ethical challenges and limitations still exist. While explainability can enhance transparency, trust, and accountability, its integration is not always straightforward or uniformly beneficial. This section addresses five key areas of concern: technical opacity, student over-reliance, educator deskilling, exacerbation of bias, and explainability trade-offs.

Technical and Pedagogical Opacity

Although XAI methods aim to clarify the reasoning behind AI outputs, their actual interpretability varies across user groups. Techniques such as SHAP, LIME, or attention maps are often intelligible to data scientists but remain inaccessible to non-technical users; particularly students and instructors without a background in machine learning (Miller, 2019). This mismatch leads to ‘explanation gaps,’ where users struggle to connect visual or statistical explanations with pedagogically meaningful insights.

Moreover, even human-readable explanations may oversimplify or misrepresent complex model behavior, risking a false sense of understanding. This challenge is severe in educational domains where learners are developing critical reasoning skills and may assume AI outputs are pedagogically sound simply because they are somehow explained.

Over-Reliance and Cognitive Offloading

Several student reflections raised concerns about dependence on AI-generated feedback, particularly when it is perceived as more immediate or objective than teacher feedback. Moreover, the research showing that students may defer too readily to algorithmic outputs, even when explanations are provided (Binns et al., 2018). Over time, such cognitive offloading may undermine the development of self-regulation, metacognition, and critical thinking, which are essential for lifelong learning (Facione, 1990; Winne & Hadwin, 1998).

Additionally, when students receive continuous AI feedback, they may become less willing to revise independently or question suggestions; treating explanations as directives rather than tools for learning. XAI systems must therefore be designed not only to inform but to encourage reflection and dialogue.

Educator De-skilling and Role Reconfiguration

While XAI can support teachers in assessment and instructional planning, there is concern that over-reliance on explainable dashboards and analytics tools may erode educators' professional judgment. In several reviewed platforms, instructors reported that they increasingly ‘monitor AI monitors’ rather than engage directly with student work prompting fears of professional deskilling (Selwyn, 2019).

Moreover, some institutions have begun to treat AI explanations as ‘second opinions’ or even as final judgments in academic disputes, marginalizing teachers' authority. This raises important questions about the distribution of epistemic responsibility in AI-supported classrooms as who ultimately decides what counts as ‘correct,’ ‘fair,’ or ‘appropriate’?

Amplifying Existing Biases Under the Veil of Transparency

A paradox of XAI is that it may create a false sense of fairness or neutrality. Just because an AI system provides explanations does not mean it is unbiased. Indeed, in several international cases, students and instructors observed that explainable models still exhibited structural biases, penalizing certain linguistic styles, dialects, or culturally specific examples, despite being ‘transparent.’

This illusion of fairness can be more dangerous than outright opacity, as it may lead to uncritical acceptance of flawed outputs. Without critical audits, explainability can serve to mask rather than mitigate discrimination, especially in high-stakes contexts such as admissions, grading, or career recommendations (Barocas et al., 2020).

Trade-offs Between Accuracy and Interpretability

Finally, a persistent limitation in XAI development is the trade-off between model performance and explainability. High-performing models, such as large language models (LLMs) or deep neural networks, often resist full transparency due to their complexity. Conversely, interpretable models (e.g., decision trees or rule-based systems) may lack the predictive accuracy or flexibility needed for nuanced educational tasks (Rudin, 2019).

Educators and developers are thus forced to navigate a design dilemma: ‘should we prioritize performance or explainability?’ Some hybrid approaches have emerged combining black-box models with post-hoc explanations or using AI as a support layer rather than a decision-maker. However, these strategies are still evolving, and no consensus exists on best practices.

These challenges suggest that explainability alone cannot guarantee ethical AI in education. Explanation alone is insufficient — stakeholders must also quantify uncertainty — then argue by analogy that educational AI systems also need mechanisms for assessing reliability, not just transparency (Salvi, et al., 2025). Besides, it must be situated within broader frameworks of pedagogical responsibility, algorithmic accountability, and human-AI collaboration. As the next section explores, advancing XAI in education requires not only technical refinement but also co-design, policy alignment, and cultural adaptation.

Design Implications and Policy Recommendations

To ensure that Explainable AI in education promotes transparency, equity, and meaningful learning outcomes, both system designers and policymakers must engage in deliberate planning and inclusive development. This section offers strategic recommendations for aligning explainability features with pedagogical goals, user needs, and institutional accountability. It also calls for a broader socio-technical ecosystem that supports ethical AI deployment across learning environments.

Design with Pedagogical Purpose

XAI in education should be developed with clear pedagogical objectives, not just technical interpretability. Explanations must go beyond computational outputs to support learner reflection, formative feedback, and conceptual understanding. For example, instead of showing feature importance scores or statistical confidence intervals, systems can use rubric-aligned justifications, natural language reasoning, or visual indicators that align with instructional goals.

Developers should involve educators and learners early in the design process to co-create explanation types that are intuitive, age-appropriate, and aligned with curriculum standards (Holstein et al., 2020). Particularly in formative settings, explanations should scaffold student self-regulation and critical engagement rather than encourage passive acceptance of AI outputs.

Salido et al. (2025) examine skills and strategies that integrate critical thinking and AI in higher education, arguing that students' capacity to question AI explanations and generate alternative interpretations should be an integral part of XAI designs.

Differentiate Explanations for Multiple Stakeholders

Effective XAI design requires audience-sensitive explanation layers. All stakeholder groups; students, teachers, administrators, and parents each need tailored insights. For instance:

- Students benefit from actionable and pedagogically contextualized feedback.
- Teachers require system transparency to verify and contest AI judgments.
- Administrators need system-level overviews for resource planning or early interventions.
- Parents (particularly in K–12) value interpretable summaries of AI-influenced decisions impacting their child's learning.

Multi-layered interfaces that offer different depths of information can meet this need. For example, a teacher dashboard might include confidence metrics and behavior traces, while a student portal offers simplified learning suggestions.

Foster Trust Through Transparency and Contestability

Beyond providing explanations, XAI systems must build user trust through contestability, user control, and error correction mechanisms. Users should be able to challenge AI outputs, provide counterexamples, or leave of automated decisions in certain high-stakes contexts. Explainability should serve not only as a post-hoc rationalization but as a tool for dialog and co-agency.

Policy frameworks should encourage educational institutions to implement AI appeal mechanisms, log user interactions with explanations, and regularly audit models for fairness and pedagogical alignment. Trustworthy XAI is not just explainable, it is also accountable.

Embed Explainability into Teacher Training

To bridge the gap between AI designers and classrooms, professional development programs must equip educators with AI literacy, including how to interpret and critique explainable models. Ho and Lee (2025) show that teachers often rely on intuitive ethical judgments when using AI tools, but that these intuitions become more coherent and consistent when supported by XAI-based explanations and clear institutional guidelines. Training should include:

- Understanding basic principles of how AI systems make decisions.
- Interpreting different forms of XAI output (e.g., rubrics, visual saliency, rule trees).
- Identifying when AI explanations might be misleading, biased, or inappropriate.

Such capacity-building empowers teachers to act as critical mediators between students and intelligent systems, fostering a culture of inquiry rather than dependence.

Establish Ethical and Regulatory Guidelines

The integration of XAI in education must be governed by ethically informed policies that safeguard student rights and educational values. Policymakers should consider:

- Mandating transparency in all AI systems used for grading, prediction, or recommendation.
- Protecting data privacy and consent, particularly in child-centered learning platforms.
- Requiring third-party audits of AI tools used in public education systems.
- Encouraging adherence to global AI ethics frameworks, such as UNESCO's Guidelines on AI in Education or the EU's AI Act.

National and institutional AI strategies must define minimum standards for explainability in education, especially when algorithmic decisions influence student placement, academic performance, or resource access. For example, Madanchian and Taherdoost (2025) systematically classify decision-making criteria for the selection of AI tools in digital education, emphasizing that institutions should develop rubrics that include transparency, explainability, and ethical compliance.

Promote Inclusive Design and Equity-Driven Innovation

Explainability must not become a luxury feature available only in well-funded schools or platforms. Equity in XAI requires universal design principles, multilingual interfaces, and accessibility features that accommodate neuro-diverse learners, language minorities, and students in low-resource environments.

Funding programs and public-private partnerships should prioritize the development of open-source, explainable AI tools that can be adapted across diverse educational settings, including in rural, under-resourced, or non-Western contexts.

Explainable AI is not a mere technical feature. It is a pedagogical, ethical, and policy commitment. Its successful implementation requires collaboration among system developers, educators, students, and regulators. Only through participatory, purpose-driven design and responsive governance can XAI truly enhance learning while safeguarding the values of fairness, transparency, and human-centered education.

The following checklist in Table 4 provides an exemplary guide for evaluating the ethical implementation of explainable AI in education. It spans key domains of system design, user experience, pedagogical alignment, governance, and inclusion. This tool is intended for educators, developers, and policymakers to ensure that XAI systems align with core educational values and support equitable, transparent learning environments.

Table 4.

Checklist for Ethical XAI Deployment in Education

Category	Checklist Item
Design & Development	Were educators and students involved in co-design of the XAI system?
Design & Development	Does the system provide multiple types of explanations (textual, visual, rubric-based)?
Design & Development	Is the explanation tailored for different stakeholders (e.g., students, teachers, parents)?
User Experience	Are explanations understandable to non-technical users?
User Experience	Can users contest, override, or request clarification on AI outputs?
Pedagogical Alignment	Do explanations support metacognitive reflection and self-regulation?
Pedagogical Alignment	Are feedback and suggestions aligned with curriculum and learning goals?
Governance & Accountability	Is the system audited regularly for fairness, bias, and accuracy?
Governance & Accountability	Are AI decisions logged and monitored for accountability?
Equity & Inclusion	Is the interface accessible to neuro-diverse and linguistically diverse learners?
Equity & Inclusion	Are low-resource institutions supported with open-source or low-cost alternatives?

Conclusion and Future Directions

As educational institutions increasingly integrate AI into teaching, learning, and assessment processes, the demand for systems that are not only efficient but also transparent, trustworthy, and pedagogically aligned has never been more urgent. XAI emerges as a crucial response to this need for bridging the gap between computational sophistication and human understanding.

This review has synthesized a growing body of research and practice surrounding XAI in education. It has demonstrated that while XAI offers promising opportunities for personalized feedback, data-informed teaching, and equitable decision-making, these benefits are contingent on thoughtful, context-sensitive design. Real-world cases from the United States, Europe, East Asia, and the Global South reveal a spectrum of approaches from post-hoc explanation panels in essay grading systems to interactive transparency modules in adaptive learning platforms. Together, they suggest that explainability is not a static feature but a dynamic educational process, shaped by institutional norms, cultural expectations, and the relational dynamics between learners and technologies.

Yet, explainability alone does not solve the deeper ethical dilemmas posed by AI in education. As discussed, challenges such as algorithmic bias, cognitive over-reliance, teacher deskilling, and false fairness persist even in systems that claim to be transparent. These tensions call for a paradigm shift: from explainability as a technical add-on to explainability as a design philosophy grounded in educational values.

Looking forward, several directions warrant further exploration:

- Human-centered evaluation frameworks are needed to assess the effectiveness of XAI not only in terms of model transparency but in terms of student learning outcomes, teacher agency, and user trust over time.
- Cross-disciplinary collaborations between AI engineers, educators, psychologists, and ethicists should inform system development, ensuring that explainability supports rather than undermines core educational goals.
- Policy development must keep pace with technological advancement, mandating minimum explainability standards in public education systems and fostering environments where algorithmic decisions can be contested, audited, and improved.
- Cultural and linguistic localization of XAI tools remains a major gap. Most current systems are developed for Anglophone contexts, leaving many learners underserved. Future work must prioritize inclusive, multilingual, and contextually adaptive explanations.
- Student voice and co-agency should be central in shaping the future of explainable systems. Learners must not only understand how AI makes decisions but also participate in the design of systems that affect their educational trajectories.

In sum, the future of XAI in education lies not in more sophisticated algorithms alone, but in cultivating transparency ecosystems where learners, educators, and systems co-construct knowledge with clarity, fairness, and shared accountability.

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Genişletilmiş Özet

Giriş

Yapay Zekâ (YZ), otomatik notlandırma ve kişiselleştirilmiş öğrenme önerilerinden öğrenci danışmanlığına ve performans tahminine kadar her şeyi etkileyen modern eğitimde dönüştürücü bir güç haline gelmiştir. Ancak, bu YZ araçları eğitimde giderek daha önemli roller üstlendikçe, şeffaf olmama, önyargılı olma ve hesap verebilirlik eksikliği konusundaki endişeler önemli ölçüde artmıştır. Bu endişeler, özellikle YZ çıktılarının öğrencilerin geleceklerini doğrudan etkilediği kabul kararları veya otomatik değerlendirmeler gibi durumlarda daha da belirginleşmektedir.

Açıklanabilir Yapay Zekâ (İngilizce kısaltmasıyla XAI), bu sorunları ele almak için hayati bir araştırma ve uygulama alanı olarak ortaya çıkmıştır. XAI, yapay zekâ sistemlerini şeffaf, yorumlanabilir ve insan kullanıcılar için hesap verebilir hale getirmek üzere tasarlanmış teknik ve ilkelerden oluşur. Eğitimde bu, teknik bir yükseltmenin ötesinde, pedagojik ve etik bir zorunluluktur. Öğrenciler, eğitimciler, veliler ve kurum liderleri gibi paydaşların, özellikle de bu çıktıların akademik ve gelişimsel sonuçları olduğunda, yapay zekâ sistemlerinin sonuçlarına nasıl ulaştığına dair görünürlüğe ihtiyaçları vardır.

Yeni alinyazın, eğitimsel yapay zekâda açıklanabilirliği adalet, güven, öğrenci özerkliği ve kapsayıcılık gibi daha geniş değerlerle ilişkilendirmektedir. Yapay zekâ araçlarının eğitimdeki faydaları yaygın olarak kabul görürken, istenmeyen önyargı, veri kötüye kullanımı ve şeffaf olmayan sistemlere güvenmenin riskleri konusundaki endişeler, daha yorumlanabilir çözümlere doğru bir yönelimi teşvik etmiştir. Bu makale, gerçek eğitim bağlamlarında XAI'yi uygulamaya yönelik küresel çabaları incelemekte ve teknik yaklaşımlarını, etik zorluklarını ve pratik politika etkilerini incelemektedir.

Teorik Arka Plan

XAI, özünde yapay zekâ çıktıları ve karar alma süreçlerini insanlar için anlaşılır kılan yöntemlerden oluşur. Geleneksel yapay zekâ sistemleri, özellikle de derin öğrenmeye dayalı olanlar, genellikle kara kutu işlevi görür ve bu çıktılara nasıl ulaştıkları konusunda net bir gösterge olmadan sonuçlar üretirler. XAI, bir kararı etkileyen özellikleri ve sistemin öngörüsüne olan güvenini ortaya çıkarmaya yardımcı olan SHAP, LIME ve belirlenlik haritaları gibi araçlar aracılığıyla şeffaflık sunarak bu durumu değiştirmeyi amaçlamaktadır.

Eğitim bağlamında açıklanabilirlik özel bir öneme sahiptir. Öğrenme, geri bildirim, gerekçelendirme ve düşünmeyle şekillenen bir süreçtir. Bilişsel öğrenme teorilerine ve yapılandırmacı pedagojilere göre, öğrenciler belirli bir sonucun neden ortaya çıktığını anlayabildiklerinde, bunu önceki bilgilerle ilişkilendirebildiklerinde ve aktif olarak anlamlandırma sürecine katılabildiklerinde en iyi şekilde öğrenirler. Etkili bir şekilde uygulandığında, Açık Öğrenme, öğrencilerin üst bilişsel ve öz düzenleme becerilerini geliştirmelerine yardımcı olarak yalnızca bir puanlama makinesi değil, aynı zamanda öğrenmeyi kolaylaştıran bir araç haline gelir.

Etik çerçeveler, açıklanabilirliğe duyulan bu ihtiyacı desteklemektedir. UNESCO ve AB'nin Yapay Zekâ Yüksek Düzeyli Uzman Grubu, özellikle sistemler çocuk merkezli eğitim ortamlarında kullanıldığında şeffaflık ve anlaşılabilirliğe vurgu yapmaktadır. Teknik olarak, açıklanabilirlik küresel olabilir (bir yapay zekâ modelinin nasıl çalıştığına dair sistem genelinde bir bakış açısı sunabilir) veya yerel olabilir (bireysel kararları açıklayabilir). Her iki form da eğitimde kritik öneme sahiptir: Öğretmenler ve politika yapımcılar küresel iç görüleri ihtiyaç duyarken, öğrenciler belirli sonuçları anlamak için yerel açıklamalara ihtiyaç duyar. Eğitimsel, etik ve bilgisayar bilimi perspektiflerinin birleşimi, Yapay Zekâyı yalnızca teknik bir özellik değil, çok boyutlu bir gereklilik olarak konumlandırır.

Açıklanabilir Yapay Zekanın Eğitimdeki Uygulamaları

XAI, değerlendirme, öğretim ve karar alma süreçlerinde şeffaflık sağlayarak eğitimde çeşitli alanlarda giderek daha fazla uygulanmaktadır.

Otomatik Deneme Puanlaması (AES): PEG ve e-rater gibi AES sistemleri, denemeleri puanlamak için makine öğrenimini kullanır, ancak şeffaflık eksikliği nedeniyle eleştirilmiştir. XAI, dil bilgisi, tutarlılık ve kelime bilgisi gibi özelliklerin puana nasıl katkıda bulunduğunu ortaya çıkararak bu sistemleri geliştirir. SHAP ve dikkat tabanlı görselleştirmeler gibi araçlar, öğrencilerin çalışmalarını daha etkili bir şekilde anlamalarına ve gözden geçirmelerine yardımcı olur. Öğretmenler ayrıca, belirli geri bildirimlerin neden verildiğini gösteren panolardan yararlanarak daha adil ve güvenilir bir notlandırma sağlar.

Uyarlanabilir Öğrenme Sistemleri: Knewton, Smart Sparrow ve ALEKS gibi platformlar, öğretimi kişiselleştirmek için yapay zekâ kullanır. Bu platformlardaki ilerleme gerekçeleri ve "bu öneri neden" düğmeleri gibi XAI özellikleri, öğrencilerin belirli içeriklerle neden karşılaştıklarını anlamalarına yardımcı olur. Öğrenciler, sistem uyarlamaları açıklandığında kendilerini daha güvende ve motive hissettiklerini bildirirler. Öğretmenler, öğrenme yolları hakkında bilgi edinerek öğretim planlamasını iyileştirir.

Dil Öğrenmede Gerçek Zamanlı Geri Bildirim: XAI, ELSA Speak, SpeakBuddy ve iFLYTEK gibi yapay zekâ destekli dil uygulamalarında çok önemlidir. Bu platformlar, ham puanların ötesine geçerek telaffuz ve dil bilgisi konusunda

ayrıntılı görsel ve metinsel geri bildirimler sunar. Örneğin, dil yerleştirme veya dil bilgisi hatalarını vurgulamak, eyleme geçirilebilir öğrenme ipuçları sağlar. Bu, yalnızca dil edinimini artırmakla kalmaz, aynı zamanda öğrencinin kaygısını azaltır ve kullanıcı memnuniyetini artırır.

Öğrenci Performans Tahmini: Tahmine dayalı analiz araçları, çeşitli göstergelere dayanarak başarısızlık veya okulu bırakma riski taşıyan öğrencileri işaretler. XAI, hangi davranışların (kaçırılan sınavlar veya düzensiz oturum açmalar gibi) uyarıya yol açtığını göstererek bu tahminlerin yorumlanabilir olmasını sağlar. Öğretmenler, açıklama katmanları içeren bu sistemlere daha fazla güvenir ve bu tür araçlar zamanında ve bilinçli müdahaleleri destekler.

Kariyer Danışmanlığı ve Rehberliği: Naviance ve CareerNeeti gibi platformlar, eğitim ve kariyer yolları önermek için yapay zekayı kullanır. Açıklanabilir arayüzler, öğrencilere ilgi alanlarına, güçlü yönlerine ve performansa göre önerilerin nasıl oluşturulduğunu gösterir. Şeffaf mantık, güvensizliği azaltır ve bilinçli karar vermeyi teşvik eder. Etkileşimli panolar ve açıklama istemleri, öğrencilerin yollarını kişiselleştirmelerine ve gerektiğinde sistem önerilerini sorgulamalarına yardımcı olur.

Eğitimde XAI'ye İlişkin Küresel Örnek Olay Çalışmaları

Amerika Birleşik Devletleri: Turnitin Draft Coach ve Gradescope gibi araçlar artık değerlendirme ölçütü bağlantılı geri bildirim ve satır içi yorumlar gibi açıklama özellikleri içeriyor. GPT-4o tabanlı yazma destek sistemleri ayrıca, değerlendirme ölçütü unsurlarının makale içeriğiyle nasıl uyumlu olduğunu gösteren görsel açıklama panelleri de sunuyor. Bu araçlar, biçimlendirici öğrenmeyi destekliyor ve yapay zekâ notlandırmasının gizemini ortadan kaldırıyor.

Hollanda: Hollanda üniversiteleri, özellikle Twente Üniversitesi, değerlendirme araçlarında görsel açıklama kartları ve özellik önem haritaları kullanmayı denedi. Öğrenciler, edilgen ses kullanımı gibi belirli hatalara bağlı olduğunda geri bildirim daha fazla güveniyor. Eğitimciler, pedagojik uyumu ve yapay zekâ çıktılarını doğrulama becerisini takdir ediyor.

Birleşik Krallık: Birleşik Krallık kurumları, öğrencilerin yazılarını yönlendirmek için Draft Coach'taki XAI özelliklerini kullanır. Dil bilgisi veya atıf kalitesiyle ilgili açıklamalar, öğrencilerin kendi kendine düzenleme yapmalarını destekler. Anketler, öğrencilerin şeffaf, beceriye özgü ve kurumsal değerlerle uyumlu geri bildirimleri tercih ettiğini göstermektedir.

Avustralya: Mark My Words gibi platformlar, açıklama katmanlarını entegre ediyor ve hatta öğrencileri geri bildirimler üzerinde eleştirel düşünmeye teşvik eden bir "Yapay Zekâ ile Aynı Fikirde Değilim" özelliği bile içeriyor. Öğrencilerle birlikte tasarlanan PolyFeed gibi araçlar, geri bildirim okuryazarlığını geliştiriyor ve yazma etkileşimini derinleştiriyor.

Çin ve Güney Kore: iFLYTEK Akıllı Öğrenme ve Riid'in Santa Tutor'ı gibi büyük ölçekli uyarlanabilir sistemler artık içerik önerilerini açıklayan XAI unsurlarını içeriyor. Bu özellikler, yapay zekayı sınıf içi eğitimle uyumlu hale getirmeye ve özellikle sınav hazırlıklarında öğrenci direncini azaltmaya yardımcı oluyor.

Brezilya: Brezilya üniversiteleri, öğrenci katılım eğilimlerini açıklamak için XAI kullanan öğrenme analitiği panolarını pilot olarak uyguluyor. Öğretmenler, performans sorunlarını somut davranışlara dayandırarak daha iyi hedeflenmiş bir destek sağlayabiliyor. Ortaklaşa tasarlanan sistemler hem öğretmen güvenini hem de pedagojik etkinliği artırıyor.

Etik Zorluklar ve Sınırlamalar

Teknik ve Pedagojik Opaklık: SHAP ve LIME gibi araçlar kullanılsa bile, açıklamalar çoğu eğitimci ve öğrenci için fazla teknik olabilir. Aşırı basitleştirilmiş açıklamalar, karmaşık yapay zekâ mantığını çarpıtma riskiyle karşı karşıyadır ve bu da hatalı çıktılara dair yanlış bir güven oluşmasına yol açar.

Aşırı Güven ve Bilişsel Yük Atma: Öğrenciler yapay zekâ geri bildirimlerine aşırı güvenebilir, bu da öz düzenleme ve eleştirel düşünme becerilerini zayıflatabilir. Bazıları açıklamaları yansıtıcı araçlar yerine yönergeler olarak görmeye başlayarak kendi karar verme yeteneklerini zayıflatabilir.

Öğretmenlerin Becerilerinin Azaltılması: Yapay zekâ sistemleri nüfuz kazandıkça, bazı öğretmenler aktif değerlendiriciler yerine izleme rollerine itilmiş hissediyor. Yapay zekâ araçlarına güvenmenin mesleki yetkiyi azalttığı ve sorumluluğu insan eğitimcilerden uzaklaştırdığı yönünde endişeler var.

Şeffaflığın Gizlediği Önyargı: Kararları açıklanabilir kılmak, önyargıyı ortadan kaldırmaz. Bazı XAI sistemleri, adil görünürken kültürel veya dilsel önyargıları yansıtmaya devam eder. Bu tarafsızlık yanılsaması, daha derin yapısal eşitsizlikleri gizleyebilir.

Doğruluk ve Yorumlanabilirlik Arasındaki Dengesizlikler: Daha şeffaf modeller genellikle kara kutu sistemlerinin tahmin gücünden yoksundur. Hibrit modeller bir uzlaşma sunar, ancak doğruluk ve netlik arasında denge kurmak, eğitimsel yapay zekâ tasarımında temel bir ikilem olmaya devam etmektedir.

Tasarım Sonuçları ve Politika Önerileri

Pedagojik Amaçlı Tasarım: Açıklamalar, istatistiksel jargon yerine doğal dil veya ölçüt tabanlı geri bildirim kullanılarak öğrenme hedefleriyle uyumlu olmalıdır. Geliştiriciler, eğitimcileri ve öğrencileri ortak tasarım süreçlerine dahil etmelidir.

Paydaşlar İçin Farklılaştırma: Paydaşların farklı türde açıklamalara ihtiyacı vardır. Öğrencilerin eyleme geçirilebilir geri bildirimlere, öğretmenlerin teşhislere, yöneticilerin sistem genelinde içgörülere ve velilerin anlaşılır özetlere ihtiyacı vardır. Çok katmanlı arayüzler bu ihtiyaçları karşılayabilir.

Güven ve Tartışmayı Teşvik Edin: XAI sistemleri, kullanıcıların yapay zekâ kararlarını sorgulamasına, itiraz etmesine veya geçersiz kılmasına olanak tanımalıdır. Politikalar, denetlenebilirliği ve yapay zekâ-kullanıcı etkileşimlerinin kaydedilmesini desteklemeli ve şeffaflığın hesap verebilirlikle bağlantılı olmasını sağlamalıdır.

XAI'da Öğretmen Eğitimi: Mesleki gelişim, yapay zeka açıklamalarının nasıl yorumlanacağını ve eleştirileceğini, sistem sınırlarının nasıl anlaşılacağını ve XAI'nın pedagojik yargıyı değiştirmek yerine bir iskele aracı olarak nasıl kullanılacağını kapsamalıdır.

Etik İlkeler ve Düzenlemeler: Hükümetler ve kurumlar, notlandırma veya yerleştirme için kullanılan yapay zekâda şeffaflığı zorunlu kılmalıdır. Veri koruma, üçüncü taraf denetimleri ve uluslararası etik kurallara (örneğin UNESCO, AB Yapay Zekâ Yasası) uyum gereklidir.

Kapsayıcı ve Eşitlikçi Tasarım: XAI, kaynakları yetersiz okullar, çok dilli öğrenciler ve farklı ihtiyaçları olan öğrenciler için erişilebilir olmalıdır. Açık kaynaklı geliştirme ve kamu-özel sektör ortaklıkları, eşit erişimi sağlamaya yardımcı olabilir.

Sonuç ve Gelecek Yönlendirmeleri

XAI artık teorik bir eklenti değil; eğitimde sorumlu yapay zekâ entegrasyonu için temel bir gerekliliktir. Bu inceleme, XAI'nin kişiselleştirilmiş öğrenmeyi, biçimlendirici değerlendirmeyi, erken müdahaleleri ve kariyer rehberliğini nasıl desteklediğini vurgulamıştır. Bölgeler arası vaka çalışmaları, farklı stratejiler ve etkiler ortaya koyarak, açıklanabilirliğin tek bir kalıba uymadığını, bağlama bağlı ve ilişkisel olduğunu ortaya koymaktadır.

Ancak XAI her yerde deva değil. Önyargı, aşırı güven ve öğretmen otoritesinin azalması gibi etik gerilimler, açıklanabilir sistemlerde bile varlığını sürdürüyor. Gelecekteki ilerleme, XAI'nin teknik bir yama olarak değil, güven, şeffaflık ve eğitimde eşitlik temelinde bir tasarım felsefesi olarak yeniden çerçevelenmesine bağlı.

Temel öneriler arasında insan merkezli değerlendirme çerçeveleri geliştirmek, disiplinlerarası iş birliğini desteklemek, duyarlı politikalar oluşturmak ve yerelleştirilmiş, kapsayıcı XAI araçlarının yaygınlaştırılması yer alıyor. En önemlisi, öğrencilerin öğrenme yolculuklarını etkileyen sistemlerin şekillendirilmesinde söz sahibi olmaları gerekiyor. XAI'nin geleceği sadece daha iyi modellerde değil, aynı zamanda öğrencilerin, öğretmenlerin ve yapay zekânın anlamlı ve adil eğitim deneyimleri birlikte yarattığı şeffaflık ekosistemlerinin oluşturulmasında yatıyor.